

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

001 This supplement records the material supporting the
002 main paper without being required to follow its argu-
003 ment: the region-partition procedure and its sensitiv-
004 ity study, complete audit tables for both tracks at ev-
005 ery horizon, measurement coverage, protocols for the
006 two complementary perceptual checks, and additional
007 qualitative material. The text-conditioned and image-
008 conditioned tracks are reported here at equal depth.
009 Section numbering is independent of the main paper.

010 1. Region Partition

011 Derivation. For every generated sequence we aver-
012 age optical-flow magnitude [7] over the leading 12%
013 of sampled frame pairs and threshold it by Otsu's
014 method [4] into a dynamic region and its complement.
015 The static region is then reduced by a fixed morpholog-
016 ical erosion and a 4% frame border discarded; together
017 these form the ignored transition band, so boundary
018 pixels—where the partition is least certain and flow
019 least reliable—contribute to neither region. The same
020 procedure applies to both tracks.

021 Why the estimate is restricted to an early window.
022 Deriving a partition from a sequence's entire motion
023 field would be circular, since accumulated background
024 drift would itself be read as evidence that the affected
025 region is dynamic. Restricting the estimate to the early
026 window breaks that circularity in the direction that
027 matters: at the 60s horizon the partition is fixed from
028 the first 7.1s, before long-horizon drift has accumu-
029 lated, and the metrics are evaluated over the remain-
030 der.

031 Limitations. The partition is obtained automatically
032 rather than by annotation, so no inter-annotator agree-
033 ment is reported, and for the image-conditioned track
034 it is derived per generated sequence rather than once
035 from the shared source image. The early window is
036 a weaker guarantee than a first-frame annotation: a
037 system that drifts within the first 12% of its rollout
038 can influence its own partition. We therefore treat
039 partitions as measured quantities and report the con-
040 sequence directly—every measurement is recomputed
041 under systematic erosion and dilation of the boundary,
042 with the response in Table 1.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

043 2. Validation Protocol

044 Geometric severity is specified in a common physical
045 unit—the mean pixel displacement a corruption in-
046 duces over the frame by the end of the rollout—and
047 each family's parameter is derived from it using the
048 frame geometry: a translation of d pixels, a rotation of
049 $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean
050 distance from the frame centre. This is what makes the
051 geometric families comparable. A parameter in degrees
052 is not commensurate with one in pixels: at the evalua-
053 tion resolution one degree of rotation induces roughly
054 2.7px of mean displacement, so families compared by
055 their native parameters differ in injected severity by
056 more than an order of magnitude, and a factor can ap-
057 pear non-monotone simply because the corruption was
058 too weak to clear the clip's own baseline drift.

059 3. Measurement Coverage

060 Coverage is counted from valid per-video measure-
061 ments rather than from the presence of a results file, so
062 a run that terminated early is reported as partial. Sep-
063 arately, fBD is feature-based [6] and declines to report
064 where too few repeatable keypoints survive; those ab-
065 stentions are tabulated rather than folded silently into
066 a smaller n .

track	horizon	measured	fBD abstained	rate
I2V	120s	400	0	0.0%
I2V	240s	65	0	0.0%
I2V	5s	180	0	0.0%
I2V	60s	630	7	1.1%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	185	0	0.0%

Table 2. Measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Every abstention in this benchmark occurs on a single desert dust-storm scene, whose flat, dust-obscured ground offers no stable structure to track, and it occurs there for every system alike—so it reflects the scene, not a method. The remaining factors are reported for those clips as usual; only fBD is withheld.

067 4. Complete Audit, Both Tracks

068 The main paper reports the text-conditioned track
069 at 60s, the tier whose sample supports method-level
070 separation. Both tracks appear here at every hori-
071 zon, with uncertainty from a percentile bootstrap over
072 items [1]. For the image-conditioned track, systems
073 marked matched run under a common long-horizon
074 wrapper rather than their released configuration, so
075 their scores are read as deployment sensitivity rather
076 than as the published method’s performance.

077 5. Whole-Frame and SNF-Bench Rank Corre- 078 lation

079 6. Deployment Sensitivity

080 7. Signed Drift Attenuation

081 Drift attenuation is stored signed and reported clipped
082 to $[0, 1]$. Negative values are informative rather than
083 anomalous: where local flow opposes the estimated
084 global field, compensation increases the measured mag-
085 nitude. We report their incidence rather than suppress-
086 ing it.

087 8. Evaluation-Set Composition

088 9. Complementary Perceptual Checks

089 Neither check is aggregated into a factor and neither is
090 a preference study. Both answer one question: does an
091 independent judge, with no access to our factors, order
092 the same clip pairs the same way?

Pair selection. Pairs are chosen by rule rather than
by eye. For each axis, candidate pairs of systems on
a shared prompt are binned by the relative gap in the
relevant factor and sampled across three bands (0.05–
0.20, 0.20–0.50, 0.50–1.00), so agreement is measured
over the whole operating range rather than only where
any metric would look good. Twelve pairs are drawn
per axis.

Human study. Raters answer exactly two forced-
choice questions—which video’s background moves
more, and in which video the moving content slows
or stops more—and are never asked which is better,
since “better” conflates the two failures the benchmark
separates. Trial order and left/right presentation are
randomised per rater, choices are resolved back to sys-
tem identities server-side, and system names are never
shown. Ties are permitted: forcing a choice on an in-
distinguishable pair manufactures noise. Twenty-four
trials per rater, about eight minutes, three raters.

Vision-language judge. The same pairs, with the
same wording, are put to an open-weight vision-
language model [5], following the practice of MLLM-
scored generation benchmarks [2, 3] but restricted to
the two axis questions rather than an overall score.
Each pair is asked twice with the clips swapped; a
judge answering by position rather than by content
disagrees with itself, so agreement is computed only
over self-consistent trials and the self-consistency rate
is reported beside it.

Reporting. Per axis we report the agreement rate
with a binomial interval, the tie rate, and agreement
resolved by metric gap. The last matters: a factor that
agrees with observers only on large gaps is a weaker re-
sult than one that agrees throughout, and pooling hides
the difference.

10. Additional Qualitative Material

References

- [1] Bradley Efron. Bootstrap methods: Another look at the jackknife. *Ann. Statist.*, 7(1):1–26, 1979. 2
- [2] Hui Han, Siyuan Li, Jiaqi Chen, Yiwen Yuan, Yuling Wu, Yufan Deng, Chak Tou Leong, Hanwen Du, Junchen Fu, Youhua Li, Jie Zhang, Chi Zhang, Li-jia Li, and Yongxin Ni. Video-Bench: Human-aligned video generation benchmark. In *CVPR*, pages 18858–18868, 2025. 2
- [3] Xuan He, Dongfu Jiang, Ge Zhang, Max Ku, Achint Soni, Sherman Siu, Haonan Chen, Abhranil Chandra, Ziyang Jiang, Aaran Arulraj, Kai Wang, Quy Duc

Method	status	setting	5s	60s	120s	240s
CausVid	public	native	V6 B	V23 T23 B	V6 B	V4 B
Self-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Infinite-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Rolling-Forcing	public	native	V6 B	V24 T24 B	V6 B	V4 B
Reward-Forcing	public	native	V12 B	V23 T23 B	V6 B	V4 B
LongLive	public	native	V6 B	V23 T23 B	V6 B	V4 B
Causal-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Method	status	setting	5s	60s	120s	240s
Causal-Forcing++ (2-step)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (1-step)	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing (framework)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (2-step, native nfpb=1)	public	native	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Self-Forcing	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
CausVid	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Wan2.1-I2V-14B-480P	public	native	V10 T10 X B	.	.	.
Wan2.2-I2V-A14B	public	native	V10 T10 X B	.	.	.
LTX-Video 13B-0.9.8-distilled	public	native	V10 T10 X B	.	.	.
Causal-Forcing++ (1-step, native)	public	native	.	V30 T30	V20 T20	.

Table 3. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

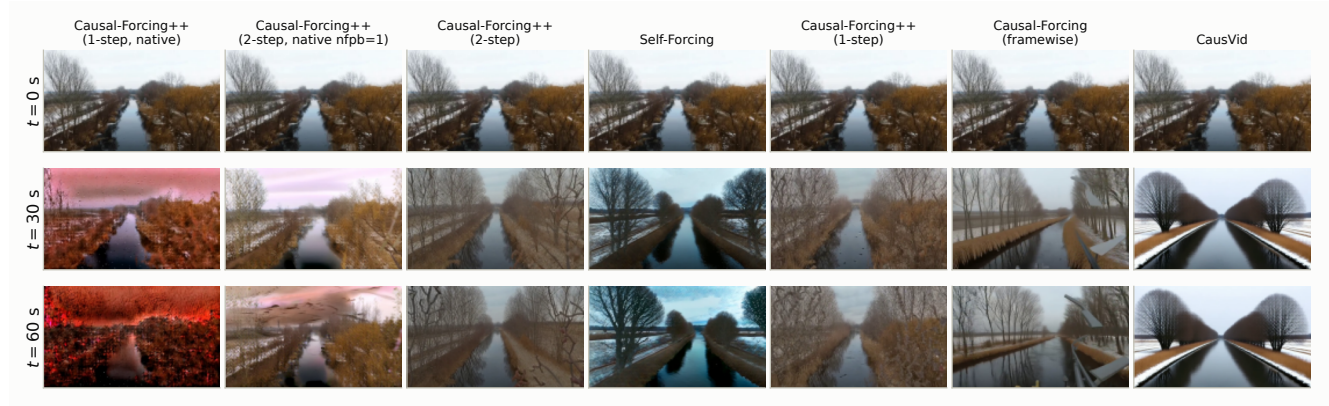


Figure 1. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Columns are ordered by increasing static-region drift.

- Do, Yuansheng Ni, Bohan Lyu, Yaswanth Narsupalli, Rongqi Fan, Zhiheng Lyu, Bill Yuchen Lin, and Wenhui Chen. VideoScore: Building automatic metrics to simulate fine-grained human feedback for video generation. In Proc. Conf. Empirical Methods Natural Language Process. (EMNLP), pages 2105–2123, 2024. 2
- [4] Nobuyuki Otsu. A threshold selection method from gray-level histograms. IEEE Trans. Syst. Man Cybern., 9(1):62–66, 1979. 1
- [5] Qwen Team. Qwen3-VL technical report. arXiv preprint arXiv:2511.21631, 2025. 2
- [6] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. ORB: An efficient alternative to SIFT or SURF. In ICCV, pages 2564–2571, 2011. 1
- [7] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs field transforms for optical flow. In ECCV, pages 402–419, 2020. 1

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 4. Complete T2V audit, all horizons. The main paper reproduces the 60s and 120s panels; the 5s tier is an initialisation check and the 240s tier is diagnostic. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	matched	3.69	10.0	–	0.65	0.88	0.67	0.22
Causal-Forcing++ (1-step)	matched	5.28	11.4	–	0.51	0.66	0.66	0.24
Causal-Forcing (framewise)	matched	16.1	45.3	–	3.28	0.40	0.68	0.12
Causal-Forcing++ (2-step, native nfpb=1)	native	1.22	10.6	–	0.83	0.49	0.68	0.11
Self-Forcing	matched	9.66	5.74	–	0.33	0.65	0.74	0.07
CausVid	matched	15.2	10.8	–	0.90	0.95	1.04	0.06
Wan2.1-I2V-14B-480P	native	5.01	21.3	–	4.84	1.03	0.52	0.16
Wan2.2-I2V-A14B	native	6.31	25.3	–	9.08	0.95	0.50	0.26
LTX-Video 13B-0.9.8-distilled	native	6.65	17.8	–	3.68	0.90	0.39	0.18
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	–	0.67	0.86	0.67	0.15
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (framewise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, native nfpb=1)	native	9.94	6.91	–	0.81	0.98	0.75	0.11
Self-Forcing	matched	17.3	5.52	–	0.74	0.88	0.67	0.14
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, native)	native	10.4	5.25	–	0.52	0.97	0.83	0.14
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	matched	9.65	8.81	–	0.93	0.91	0.58	0.14
Causal-Forcing++ (1-step)	matched	7.26	9.39	–	0.83	0.84	0.62	0.16
Causal-Forcing (framewise)	matched	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, native nfpb=1)	native	11.4	17.3	–	1.54	0.96	0.75	0.06
Self-Forcing	matched	23.0	10.3	–	0.78	0.44	0.66	0.20
CausVid	matched	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, native)	native	13.6	7.39	–	0.43	0.68	0.77	0.20
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	matched	10.4	4.83	–	0.63	1.03	0.55	0.09
Causal-Forcing++ (1-step)	matched	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (framewise)	matched	19.8	31.5	–	2.86	0.76	0.69	0.33
Causal-Forcing++ (2-step, native nfpb=1)	native	6.32	3.95	–	0.69	0.66	0.83	0.01
Self-Forcing	matched	22.3	5.74	–	1.66	0.79	0.71	0.05
CausVid	matched	18.4	2.08	–	0.83	0.83	0.44	0.07

Table 5. Complete I2V audit, all horizons. The main paper reproduces the 60 s and 120 s panels; the 5 s tier is an initialisation check and the 240 s tier is diagnostic. n is stated per horizon and is common to every row of that panel.

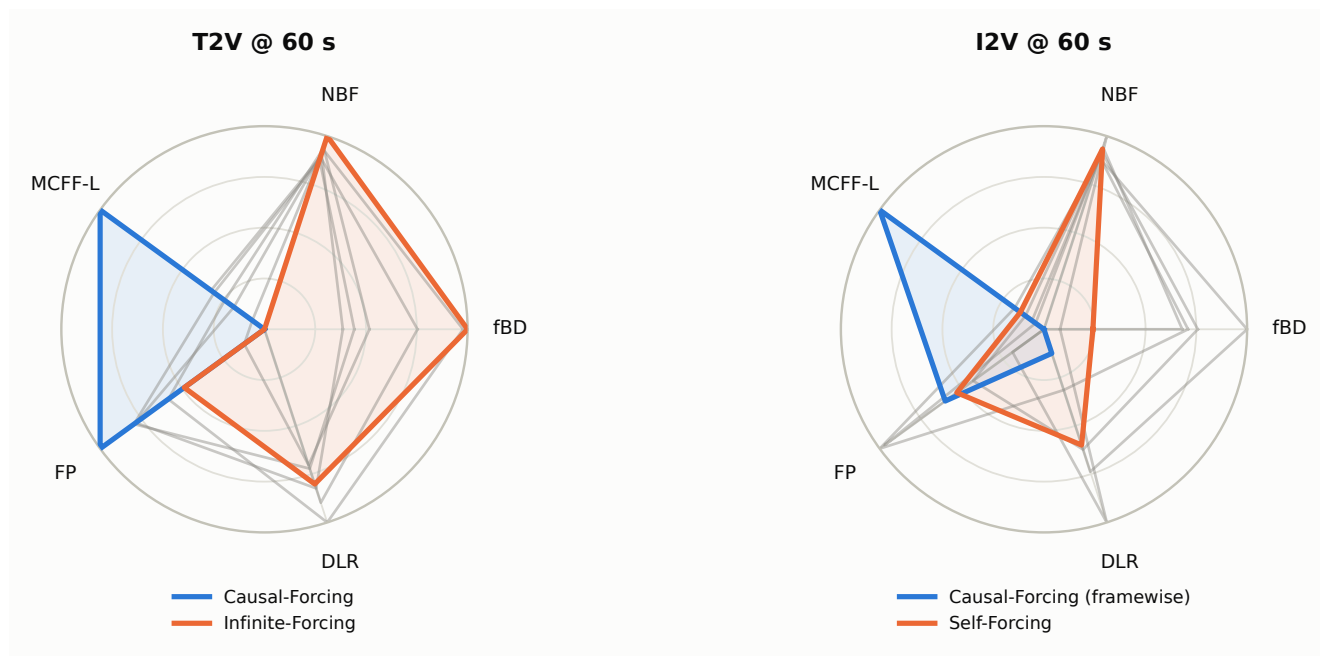


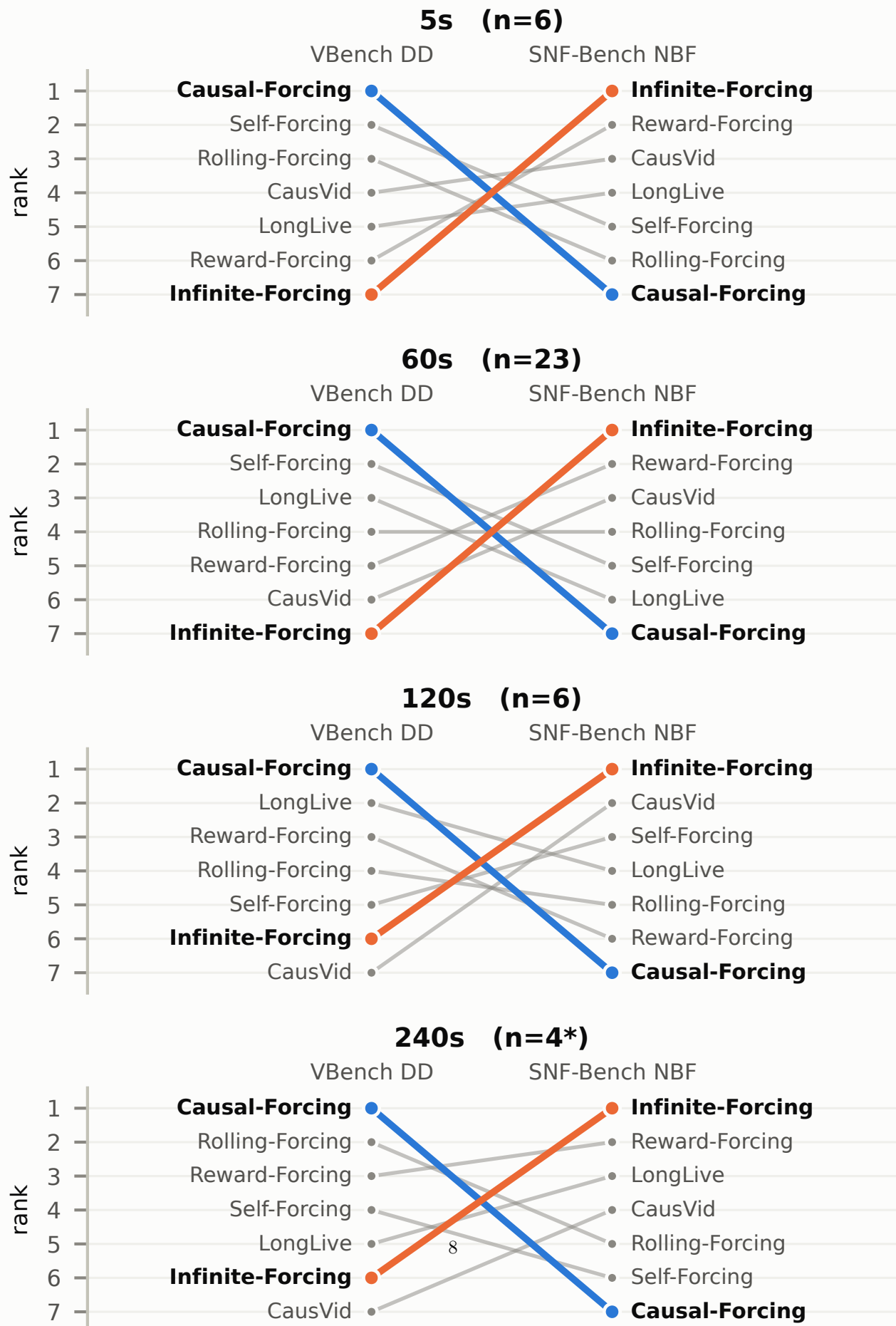
Figure 2. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile.



Figure 3. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.



Figure 4. Every audited text-conditioned system on one prompt. Columns are ordered by increasing static-region drift; the two highlighted systems are the pair a whole-frame motion score cannot separate.



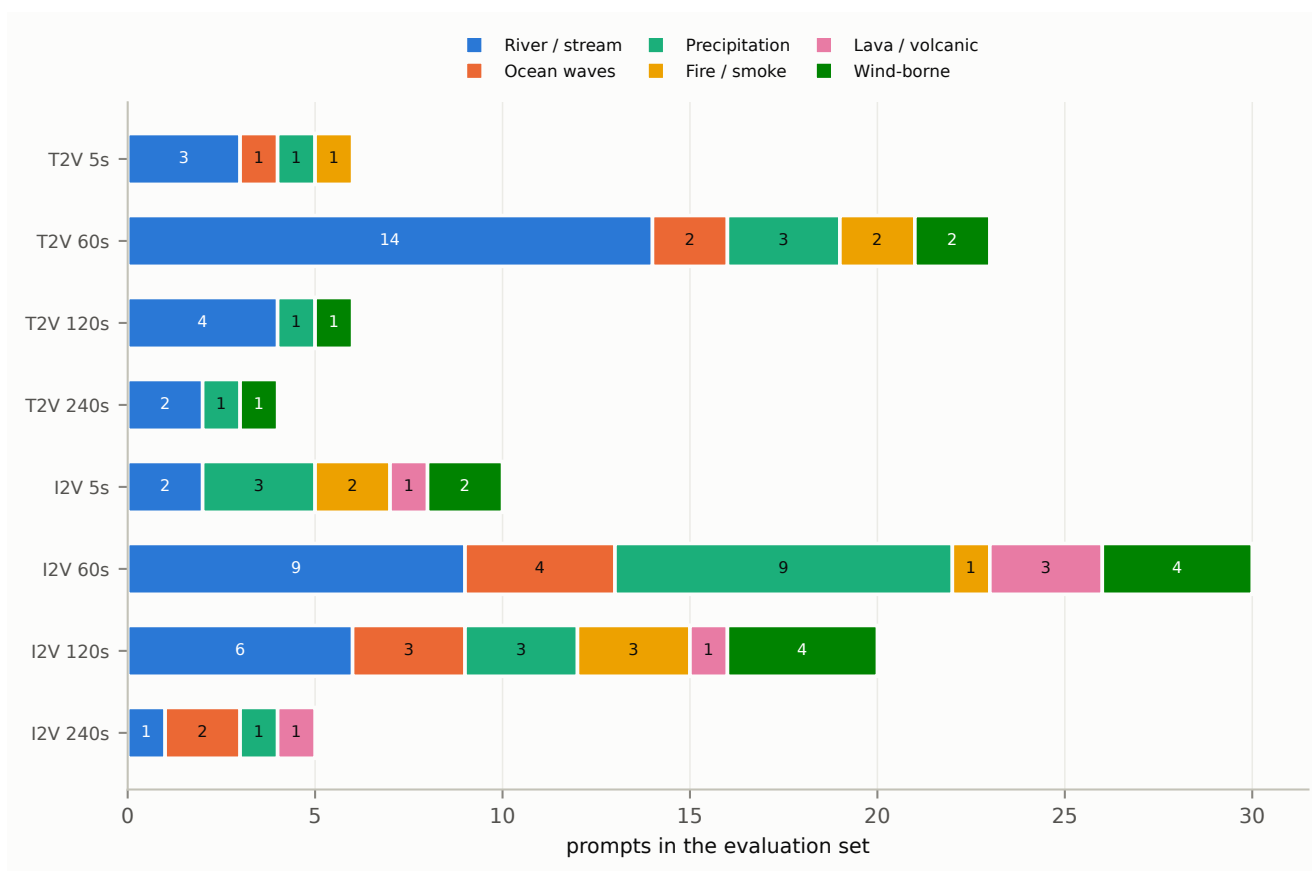


Figure 6. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth	-0.96	-0.82	-0.61	-0.64
VB-flick	-0.96	-0.82	-0.61	-0.64

Table 6. Rank disagreement at 60s. Spearman correlation between generic whole-frame metrics and SNF-Bench factors over method-level means. A positive correlation between apparent motion and background flow is the effect the benchmark isolates.

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 7. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.18	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.15	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.61	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.06	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.41	-0.14

Table 8. Deployment sensitivity. Change from each checkpoint’s native configuration to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	240s	5	0	0.0%	0.000
CausVid	i2v	5s	10	6	60.0%	-0.450
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.012
Causal-Forcing++ (1-step)	i2v	240s	5	0	0.0%	0.175
Causal-Forcing++ (1-step)	i2v	5s	10	0	0.0%	0.041
Causal-Forcing++ (1-step)	i2v	60s	30	0	0.0%	0.013
Causal-Forcing (framework)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing (framework)	i2v	240s	5	0	0.0%	0.046
Causal-Forcing (framework)	i2v	5s	10	2	20.0%	-0.043
Causal-Forcing (framework)	i2v	60s	30	0	0.0%	0.008
Causal-Forcing++ (2-step)	i2v	120s	20	0	0.0%	0.008
Causal-Forcing++ (2-step)	i2v	240s	5	0	0.0%	0.053
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.102
Causal-Forcing++ (2-step)	i2v	60s	30	0	0.0%	0.005
Causal-Forcing++ (1-step, native)	i2v	120s	20	1	5.0%	-0.062
Causal-Forcing++ (1-step, native)	i2v	60s	30	3	10.0%	-0.770
Causal-Forcing++ (2-step, native nfpb=1)	i2v	120s	20	3	15.0%	-0.192
Causal-Forcing++ (2-step, native nfpb=1)	i2v	240s	5	2	40.0%	-0.164
Causal-Forcing++ (2-step, native nfpb=1)	i2v	5s	10	1	10.0%	-0.133
Causal-Forcing++ (2-step, native nfpb=1)	i2v	60s	30	2	6.7%	-0.014
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	4	40.0%	-0.031
Self-Forcing	i2v	120s	20	3	15.0%	-0.131
Self-Forcing	i2v	240s	5	2	40.0%	-0.012
Self-Forcing	i2v	5s	10	2	20.0%	-0.026
Self-Forcing	i2v	60s	30	8	26.7%	-0.569
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.242
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.490
CausVid	t2v	120s	6	0	0.0%	0.047
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
CausVid	t2v	60s	23	0	0.0%	0.005
Causal-Forcing	t2v	120s	6	0	0.0%	0.175
Causal-Forcing	t2v	240s	4	0	0.0%	0.158
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	120s	6	0	0.0%	0.023
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	5s	6	0	0.0%	0.031
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	240s	4	0	0.0%	0.022
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	240s	4	0	0.0%	0.002
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	240s	4	0	0.0%	0.043
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table 9. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow.

category	definition
river_stream	channel water: rivers, streams, creeks, canals, waterfalls, rapids, floods
ocean_waves	open water with wave action: surf, tides, coastal swell, seascapes
precipitation	falling water or ice: rain, snowfall, and the surfaces they wet
fire_smoke	combustion plumes: wildfire, campfire, hearth, urban fire, grill smoke
lava_volcanic	volcanic flow and ejecta: lava channels, ash columns, eruption smoke
windborne	air-carried particulates and wind-driven vegetation: dust, sand, blossom, leaves, thrashing trees, drifting cloud
track/duration	n
i2v/5s	10
i2v/60s	30
i2v/120s	20
i2v/240s	5
t2v/5s	6
t2v/60s	23
t2v/120s	6
t2v/240s	4
track	source
i2v	meta
t2v	keyword
t2v	keyword-body
t2v	override
track	duration
t2v	60s

Table 10. Scene-category balance of the evaluation set, by track and horizon.